

Nonlinear Time Series Models

ARCH | GARCH | Bilinear (BAR) | Stochastic Volatility | NAR | NARX

1. The ARCH Model (AutoRegressive Conditional Heteroscedasticity)

Introduced by Engle (1982), the ARCH model was developed to capture a fundamental feature of financial time series: volatility is not constant over time. Periods of high volatility tend to cluster together, as do periods of calm. The ARCH model captures this by making the conditional variance depend on past squared values of the series.

1.1 Definition — ARCH(q)

A time series $\{X_t\}$ follows an ARCH(q) model if:

$$X_t = \sigma_t \epsilon_t, \quad \text{where } \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \alpha_2 X_{t-2}^2 + \dots + \alpha_q X_{t-q}^2$$

- ϵ_t is an i.i.d. sequence with mean 0 and variance 1 (typically standard normal or t-distributed).
- $\alpha_0 > 0$ and $\alpha_i \geq 0$ for all $i \geq 1$ — ensures variance is always positive.
- σ_t^2 is the conditional variance of X_t given past information — it is measurable with respect to the past.

1.2 Features of the ARCH Model

The ARCH model is characterised by several distinctive statistical features:

- **Volatility Clustering:** Large shocks (positive or negative) tend to be followed by more large shocks. If $|X_{t-1}|$ is large, σ_t^2 increases, raising the probability of another large $|X_t|$.
- **Leptokurtosis (Heavy Tails):** The unconditional distribution of X_t has heavier tails than a Gaussian — consistent with observed financial return distributions. Even if ϵ_t is normal, X_t is not normal.
- **Time-Varying Conditional Variance:** Unlike linear models where variance is constant, ARCH allows the variance to change with the information set at each time step.
- **Uncorrelated but Dependent:** X_t is serially uncorrelated ($E[X_t X_{t-k}] = 0$ for $k \neq 0$), yet the series $\{X_t^2\}$ exhibits significant autocorrelation — revealing non-linear dependence.
- **Mean Zero:** The conditional mean $E[X_t | \text{past}] = 0$, since $E[\epsilon_t] = 0$ and σ_t is measurable on the past.

Exam Tip: ARCH captures the 'quiet follows quiet, turbulent follows turbulent' phenomenon observed in stock markets. Identifying ARCH effects: test whether the squared residuals of a linear model show autocorrelation (Ljung-Box test on X_t^2).

1.3 Existence of a Strictly Stationary Solution — ARCH(1)

For the ARCH(1) model: $X_t = \sigma_t \epsilon_t$, $\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$, the fundamental stationarity question is: under what conditions does a strictly stationary solution exist?

Theorem (Strict Stationarity of ARCH(1)): A strictly stationary solution to the ARCH(1) model exists if and only if $E[\log(\alpha_1 \epsilon_t^2)] < 0$.

This condition is the top Lyapunov exponent condition. For the simpler case where $\epsilon_t \sim N(0,1)$:

- **Sufficient condition:** $\alpha_1 < 1$ ensures a weakly (covariance) stationary solution exists.
- **When $\alpha_1 < 1$:** The unconditional variance exists and equals $\text{Var}(X_t) = \alpha_0 / (1 - \alpha_1)$.
- **When $\alpha_1 \geq 1$:** The variance explodes — shocks do not die out and the series is non-stationary.
- **Strict stationarity:** Can exist even when $\alpha_1 \geq 1$ (i.e., even without finite variance), provided $E[\log(\alpha_1 \epsilon_t^2)] < 0$.

Economically: if $\alpha_1 < 1$, a large shock gradually fades and the market stabilises. If $\alpha_1 \geq 1$, each shock amplifies the next, causing an explosive, non-stationary process.

2. The GARCH Model (Generalised ARCH)

Proposed by Bollerslev (1986), GARCH generalises ARCH by allowing the conditional variance to depend on both past squared observations AND past conditional variances. This gives the model 'memory' — volatility from several periods ago continues to influence today's variance.

2.1 Definition — GARCH(p, q)

$$X_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \sum_i \alpha_i X_{t-i}^2 + \sum_j \beta_j \sigma_{t-j}^2$$

The most widely used specification in practice is GARCH(1,1):

$$\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

- $\alpha_0 > 0$ — baseline (long-run) variance level.
- $\alpha_1 \geq 0$ — weight given to last period's squared shock (reaction to news).
- $\beta_1 \geq 0$ — weight given to last period's conditional variance (persistence).
- ϵ_t — i.i.d. with mean 0, variance 1.

2.2 Interpretation of GARCH(1,1)

$\alpha_1 X_{t-1}^2$ (Today's Shock)	$\beta_1 \sigma_{t-1}^2$ (Volatility Memory)
○ Impact of yesterday's surprise ○ Captures immediate reaction to news ○ High $\alpha_1 \rightarrow$ sharp, reactive volatility	○ Persistence of past variance ○ High $\beta_1 \rightarrow$ volatility lingers long ○ Models prolonged post-crisis instability

After a financial crisis, markets remain unstable for days or weeks. GARCH(1,1) captures this 'slow decay' by feeding past conditional variance back into the model. ARCH alone cannot do this efficiently without very high lag orders.

2.3 Existence of a Stationary Solution for GARCH(1,1)

Theorem (Weak Stationarity): A weakly (covariance) stationary solution to GARCH(1,1) exists if and only if: $\alpha_1 + \beta_1 < 1$

- **If $\alpha_1 + \beta_1 < 1$:** The unconditional variance is finite and equals $\sigma^2 = \alpha_0 / (1 - \alpha_1 - \beta_1)$.
- **If $\alpha_1 + \beta_1 = 1$:** The model is called Integrated GARCH (IGARCH) — variance shocks are permanent (unit root in variance). The unconditional variance is infinite.
- **If $\alpha_1 + \beta_1 > 1$:** Explosive variance — the process is non-stationary and variance grows without bound.
- **Strict stationarity:** A strictly stationary solution exists under the weaker condition $E[\log(\alpha_1 \epsilon_t^2 + \beta_1)] < 0$, even when $\alpha_1 + \beta_1 \geq 1$.

Practically: GARCH(1,1) with $\alpha_1 + \beta_1$ close to (but below) 1 implies high persistence — shocks die out slowly. This matches empirical findings in equity and forex markets where volatility is highly persistent.

Exam Tip: ARCH(1) stationarity $\rightarrow \alpha_1 < 1$. GARCH(1,1) stationarity $\rightarrow \alpha_1 + \beta_1 < 1$. GARCH generalises ARCH: $\text{GARCH}(1,0) = \text{ARCH}(1)$. Higher β_1 means longer volatility memory — crucial for modelling financial crises.

3. Bilinear Models — Bilinear Autoregressive (BAR)

Bilinear models, introduced by Granger & Andersen (1978), are one of the simplest forms of nonlinear time series. They extend the linear ARMA framework by adding interaction terms between past values and past innovations (shocks). These interaction terms allow the effect of a shock to depend on the current level of the series — something linear models cannot capture.

3.1 The Bilinear Autoregressive (BAR) Model

The general bilinear model BL(p, q, P, Q) is defined as:

$$X_t = \sum a_i X_{t-i} + \sum b_{k1} X_{t-k} \epsilon_{t-1} + \epsilon_t$$

The simplest and most studied case is the Bilinear Autoregressive model BAR(1,1) or BL(1,0,1,1):

$$X_t = a X_{t-1} + b X_{t-1} \epsilon_{t-1} + \epsilon_t$$

- $a X_{t-1}$: Standard linear autoregressive component.
- $b X_{t-1} \epsilon_{t-1}$: Interaction (bilinear) term — the key nonlinear component.
- ϵ_t : i.i.d. noise with mean 0, variance σ^2 .

3.2 The Interaction Term — Economic Intuition

The term $b X_{t-1} \epsilon_{t-1}$ captures multiplicative interaction between past state and past shock. This means the impact of a shock on future values is not constant — it depends on the current level of the series:

- Strong economy + sudden policy shock → amplified effect.
- Weak economy + same shock → smaller effect.
- This explains asymmetric market responses not possible with linear ARMA.

3.3 Properties of BAR Models

- **Non-linear Structure:** The conditional mean $E[X_t | X_{t-1}, \epsilon_{t-1}] = aX_{t-1} + bX_{t-1}\epsilon_{t-1}$ is linear in X_{t-1} but nonlinear in the joint pair $(X_{t-1}, \epsilon_{t-1})$.
- **State-Dependent Variance:** $\text{Var}(X_t | \text{past}) = \sigma^2(1 + b^2X_{t-1}^2)$ — variance itself depends on the level of the process. This creates heteroscedastic behaviour.
- **Stationarity:** Requires conditions on the spectral radius of associated matrices. For BAR(1,1), a sufficient condition for stationarity involves $E[\log|a + b\epsilon_t|] < 0$.
- **Moment Conditions:** Second-order stationarity requires $a^2 + b^2\sigma^2 < 1$.

Exam Tip: The key distinguishing feature of bilinear models is the product term $X_{t-k} \cdot \epsilon_{t-l}$. This is what makes them 'bilinear' — linear in X_t and linear in ϵ_t separately, but not jointly. They are the simplest models that capture amplitude-dependent noise effects.

4. Stochastic Volatility (SV) Models

Stochastic Volatility models take a fundamentally different approach to time-varying variance compared to ARCH/GARCH. Instead of making volatility a deterministic function of past data, SV models treat volatility itself as an unobserved (latent) random process. This makes SV more flexible but also more difficult to estimate.

4.1 Model Specification

$$X_t = \sigma_t \epsilon_t, \quad \text{where} \quad \log(\sigma_t^2) = h_t$$

$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t$$

- $h_t = \log(\sigma_t^2)$: The log-volatility follows an AR(1) process.
- μ : Long-run mean of log-volatility.
- ϕ : Persistence parameter ($|\phi| < 1$ for stationarity).
- η_t : Innovation in the log-volatility process, i.i.d. $N(0, \sigma_\eta^2)$.
- ϵ_t : Return innovation, i.i.d. $N(0,1)$, typically independent of η_t .

4.2 Key Features and Comparison with GARCH

SV Model	GARCH Model
○ Volatility is a latent stochastic process ○ Cannot be computed directly from data ○ Requires Kalman filter, MCMC, or particle filter for estimation	○ Volatility is a deterministic function of past data ○ σ_t^2 can be computed exactly from observations ○ Estimated by Maximum Likelihood (fast)

- More flexible — two separate noise sources
- Naturally handles leverage effects

- Simpler but less flexible
- Leverage needs extensions (EGARCH, GJR)

- **Stationarity:** Requires $|\phi| < 1$ — the log-volatility AR process must be stationary.
- **Interpretation:** Market fear and sentiment (which drive volatility spikes) are inherently unobservable — SV models this hidden uncertainty directly.
- **Applications:** Option pricing, risk management, modelling fear indices (VIX), and regimes where sudden unobservable sentiment shifts cause volatility spikes (e.g., geopolitical events).

Exam Tip: The key difference — GARCH volatility is observable and deterministic given the past. SV volatility is latent and has its own randomness (η_t). SV is more realistic but harder to estimate. Stationarity condition: $|\phi| < 1$.

5. Nonlinear Autoregressive Models (NAR and NARX)

5.1 NAR — Nonlinear Autoregressive Model

The NAR model generalises the linear AR model by replacing the linear combination of past values with an arbitrary nonlinear function f :

$$X_t = f(X_{t-1}, X_{t-2}, \dots, X_{t-p}) + \epsilon_t$$

- **$f(\cdot)$:** A nonlinear function — could be a neural network, polynomial, sigmoid, piecewise linear (threshold), or kernel function.
- **ϵ_t :** i.i.d. noise with mean 0 and constant variance σ^2 .
- **p :** The lag order (number of past values used).

NAR is the most general framework for nonlinear univariate time series. The choice of f determines the specific model type:

- **Polynomial NAR:** f includes cross-products of lag terms — captures smooth nonlinearities.
- **Threshold AR (TAR):** f is piecewise linear, switching regimes based on a threshold variable.
- **Neural Network AR (NNAR):** f is a feedforward neural network — can approximate any continuous function.

5.2 NARX — Nonlinear Autoregressive with Exogenous Inputs

NARX extends NAR by incorporating external (exogenous) input variables U_t alongside the lagged outputs:

$$X_t = f(X_{t-1}, \dots, X_{t-p}, U_{t-1}, \dots, U_{t-r}) + \epsilon_t$$

- **U_t :** External input variable (e.g., temperature, policy rate, rainfall, price).
- **r :** Lag order for the external input.
- **$f(\cdot)$:** Nonlinear function — same options as NAR (neural network, polynomial, etc.).

NARX is the nonlinear equivalent of ARX (AR with exogenous inputs) and ARMAX. It enables the model to use both the history of the target series and relevant external drivers:

- **Electricity Demand:** X_t (demand) depends nonlinearly on past demand AND temperature U_{t-1} . A hot day causes a disproportionate surge in AC usage.
- **Macroeconomics:** GDP depends nonlinearly on its own history AND monetary policy inputs.
- **Epidemiology:** Infection rate depends on past infections AND vaccination rates.

5.3 Properties and Stationarity

- **Stationarity of NAR/NARX:** Depends heavily on the specific form of f . For f Lipschitz continuous with Lipschitz constant $L < 1$, geometric ergodicity (a strong stationarity concept) can be established.
- **Estimation:** Parameters of f are estimated by Nonlinear Least Squares (NLS), maximum likelihood, or neural network backpropagation, depending on the form of f .
- **Advantage over linear AR:** Captures threshold effects, regime changes, saturation effects, and asymmetric responses — all common in real-world data.
- **Model Selection:** Lag orders p and r selected by information criteria (AIC/BIC). The form of f is chosen based on domain knowledge or via cross-validation.

Exam Tip: NAR = nonlinear f of past outputs only. NARX = nonlinear f of past outputs + past external inputs. The key advantage is that real-world relationships are rarely linear — temperature's effect on energy demand is nonlinear; NARX captures this naturally.

6. Comparative Summary of All Models

Model	Core Formula	Key Idea	Stationarity	Use Case
ARCH(1)	$\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$	Variance driven by past shocks	$\alpha_1 < 1$	Asset volatility
GARCH(1,1)	$\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2$	Variance has memory (shocks + past variance)	$\alpha_1 + \beta_1 < 1$	Financial crises
BAR(1,1)	$X_t = aX_{t-1} + bX_{t-1}\epsilon_{t-1} + \epsilon_t$	Value $\times$ shock interaction term	Spectral cond.	Non-linear econ.
SV	$\log(\sigma_t^2) = \text{AR}(1) \text{ process}$	Hidden (latent) stochastic volatility	AR root inside unit disk	Sentiment/fear
NAR	$X_t = f(X_{t-1}, \dots) + \epsilon_t$	Nonlinear function of past values	Model-dependent	Weather, ecology
NARX	$X_t = f(X_{t-1}, U_{t-1}) + \epsilon_t$	NAR + external exogenous input	Model-dependent	Energy demand

7. Key Takeaways for the Exam

- **ARCH:** Conditional variance depends on past squared values. Features: volatility clustering, heavy tails, uncorrelated but dependent. Stationarity: $\alpha_1 < 1$ (weak); $E[\log(\alpha_1 \epsilon_t^2)] < 0$ (strict).

- **GARCH(1,1):** Extends ARCH by including past conditional variance ($\beta_1 \sigma_{t-1}^2$). Captures long memory in volatility. Stationarity: $\alpha_1 + \beta_1 < 1$.
 - **BAR:** Bilinear term $bX_{t-1}\epsilon_{t-1}$ captures amplitude-dependent, state-dependent noise effects. Simplest nonlinear model with interaction between state and shock.
 - **SV:** Volatility is a latent AR(1) process on log scale. More flexible than GARCH — two separate noise sources. Hard to estimate (MCMC/particle filter needed). Stationarity: $|\phi| < 1$.
 - **NAR:** Generalises linear AR with a nonlinear function f of past values. Covers threshold, neural network, polynomial AR as special cases.
 - **NARX:** Adds exogenous inputs to NAR. Essential when external drivers (temperature, policy) nonlinearly affect the series.
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